

Assignment 3: Generic Sorted Lists & Java Collections

Assignment 3 revisits and extends all you have learnt for Stacks, by challenging you to develop implementations for an interface of another main abstract data type—Lists:

- in **Assignment 3a** (Friday, February 13) you will use arrays to implement the essential functionality of a generic `List<E>`, as defined in the interface provided to you;
- in **Assignment 3b** (Tuesday, February 17) you will implement an algorithm to sort the elements inside the list, plus the data type `SortedList<E>` which is not quite the same;
- in **Assignment 3c** (Friday, February 20) you will extend the GUI that operates with these implementations, using Java Collections to add useful functionality.

These assignments are incremental, meaning that they build on each other. You start by downloading the IntelliJ/Maven project `assignment3.v1.1.0.zip` from DTU Learn. Assignment 3a begins from those base files; once you have finished and presented your solution, Assignment 3b continues from that solution; ditto Assignment 3c.

The final complete solution to Assignment 3 will be the project that your group submits for Part 1 (out of 4) of the course—see [Submission of projects in Exam and Submissions](#) (DTU Learn).

Assignment 3a: Generic Lists

In Assignment 3a you must provide an implementation for the generic interface `List<E>`, by implementing all necessary methods in `implementations/ArrayList.java`:

0. Before you start with this assignment, a member of your group must have presented and have approved all previous assignments of the course, viz. assignments 1, 2a, and 2b.
1. **Download** `assignment3.v1.1.0.zip` from DTU Learn to your computer. Unpack this zip-file to some new sub-folder on your computer—but remember, not a folder shared in the cloud!—and open it in IntelliJ via its `pom.xml` file.
2. **Read** the documented comments for the methods in the `List<E>` generic interface, that you find in `types/List.java`, to understand what is expected of each method in terms of both normal functionality, and exceptions to be thrown or caught.
3. **Implement** the generic class `ArrayList<E>` in `implementations/ArrayList.java`, making sure that you are complying to the aforementioned documentation.
 - All the methods that must be implemented are commented with the following:
`// TODO needs implementation (Assignment 3a).`
 - You needn't implement methods that are commented with `(Assignment 3b)`. Those are intended for—you guessed it!—the next assignment.
 - Test your implementation incrementally by hand, running `uses/PersonApp.java` and interacting with the GUI for the (unsorted) `List` implementation.

4. **Test** your solution for Assignment 3a: it can be considered sufficient for presentation to a teacher or TA only after all tests in `TestArrayLists` are passing successfully.
 - Note that you needn't pass the other tests yet, e.g. `TestSortedArrayLists`, as these are intended for the next assignments.
5. **Present** your solution to the teachers or teaching assistants during the tutorial sessions on Tuesdays or Fridays, at latest on Tuesday, February 17.
 - *If you can* (IDK, can you?) you're advised to present on Friday, February 13, since the next assignments depend on this solution. If you cannot, don't worry, just present on Tuesday, February 17.
6. If you should have time left *after presenting Assignment 3a* on Feb 13, you can already start implementing the `sort(Comparator<? super E> c)` method for `ArrayList<E>`.

References and further reading

[02324 f26 L03, L04] Carlos E. Budde and Ekkart Kindler: Lecture Notes for the course Advanced programming. Spring 2026 (DTU Learn).

[Eck 2022] David J. Eck: Introduction to Programming Using Java. Version 9.0, JavaFX Edition, May, 2022. Section 10.2: Lists and Sets.